



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

End-Spring Semester Examination 2022-23

Date of Examination: 20.04.23 Session: AN Duration: 3 hrs Full Marks: 60

Subject No.: AI60004 Subject: Big Data processing Dept: Centre of Excellence in Artificial Intelligence

Specific charts, graph paper, log book required: NO

(Important Note: Efficient algorithms will be given higher credit.)

✓ Choose the best option for the following questions:

(5 x 2)

- a) Which of the operations from among `reduceByKey()`, `combineByKey()`, and `lookup()` benefit from RDD partitioning?
i) `reduceByKey()`, ii) `combineByKey()`, iii) `lookup()`, iv) ✓ all of them *Greater the number of partitions, greater is the parallelization.*
- b) You have a 4 years old cluster and the lifetime of the nodes are 5 years. You are performing k-means algorithms on a set of vectors. What would be most appropriate and reliable storage level for an RDD that contains the vectors?
i) `MEMORY_ONLY`, ii) `MEMORY_AND_DISK`, iii) `MEMORY_ONLY_2`, iv) ✓ `MEMORY_AND_DISK_2`
- c) In value-to-key design pattern, what happens if custom partitioner is not defined for composite keys in mapreduce (MR)?
i) MR runs out of memory, ii) MR automatically finds the right partitioner, iii) ✓ values for the same key may not come to the same reducer, iv) all data come to a single reducer
- d) In node embedding using random walk, what is the recommended relationship between the size of the negative sample (say s) and the degree of the node (say d)? ($k > 0$ is a constant).
i) $s = d$, ii) ✓ $s = kd$, iii) $s = k/d$, iv) $s = d^k$
- e) An RDD named R contains the elements 1,2,3,3. What does `R.filter(x => x != 1).distinct()` produce?
i) 2, ii) 1, iii) ✓ 2,3, iv) 4

(4 x 2.5)

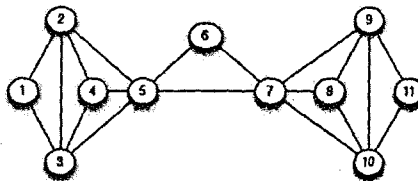
2. a) ✓ Karger's mincut algorithm is applied on a complete graph (unweighted and undirected) with n nodes. At second iteration (when picking the second edge for contraction), Find the ratio of the most probable edge to the least probable edge based on their chance of getting picked?

- b) ✓ Consider two rdds $R1 = \{(1,2), (2,3)\}$ and $R2 = \{(3,4)\}$. What will be the output of the following code?
`R1.subtractByKey(R2).reduceByKey((x, y) => x + y)`

- c) ✓ An RDD R takes 10 GB of total storage. R is persisted using the command: `R.persist(StorageLevel.MEMORY_AND_DISK_2)`. The available main memory of the cluster for storing RDD is 6 GB. How much disk this operation is going to consume?

- d) ✓ An iterative spark code performs only two operations in every iteration: The first is `map()` and the next is `groupByKey()`. If the number of iterations is set to 10, how many times the data will be shuffled?

3. a) ✓ For the graph given below, compute the betweenness centrality of the edges (8, 10) and (4, 5). (2 x 5)



- b) ✓ An undirected unweighted graph has n nodes and k edges. If m ($m < k$) edges are selected at random from the graph, what is the expected number of triangles formed by those selected edges? (No credit will be given without detailed proof.)

4. Write map-reduce algorithms for the following two problems. Write only map and reduce functions in pseudocode with clearly defined input and output of the map and reduce functions. (2 x 5)

- a) ✓ Write map and reduce function for multiplying a matrix M ($m \times n$) with an n dimensional vector v . Assume that the vector v can't be stored in any datanode's memory.

- b) You are given an undirected unweighted graph G in the form of a two column file. Each line of the file represents an edge. For example, a line containing (2 5) means there is an edge between the node 2 and 5. All the nodes are integer. Your goal is to find, for every pair of nodes having an edge, the number of common neighbors. For example, if (2 5) is an edge in the graph and 2 has neighbors (1,4,5) and 5 has neighbors (1,2,4,7) then the number of common neighbors for 2 and 5 is 2. Write a map-reduce algorithm for finding the number of common neighbours for each node pair. Note that the reduce function will always buffer the list of values for a key in main memory. Thus, the algorithm must run for all types of data distribution.

5. a) An undirected unweighted graph has n nodes and $m \geq n$ edges. The nodes are numbered $1, 2, \dots, n$. A node is called a heavy hitter node if its degree is at least $\sqrt{m}$. Prove that the maximum number of heavy hitter node in the graph is $2\sqrt{m}$. What is the time complexity of the algorithm for finding the heavy hitter nodes? (2 x 5)

b) Write down the main steps of an RDD processing? When do you want to persist an RDD in main memory?

6. a) Consider the following spark code snippet. Draw the RDD operator graph with appropriate labels to the nodes and edges. (2 x 5)

```
val lines = sc.textFile("data-file")
val errors = lines.filter(_.startsWith("ERROR"))
val messages = errors.map(_.split("\t")).map(r => r(1))
messages.cache()
```

b) Let $R1$ and $R2$ be two paired RDDs where $R1 = \{(1,2), (1,7), (2,8), (3,7)\}$ and $R2 = \{(1,5), (2,6)\}$. What would be output of the following spark codes: i) $R1.cogroup(R2).count()$ ii) $R1.join(R2)$